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# Drift $\neq$ Damage: Representation Change Does Not Reliably Identify When Encoder Adaptation Helps

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## Abstract

Continual- and transfer-learning practice reads the large representational change that fine-tuning a time-series foundation model (TSFM) induces as forgetting or damage. We show this reflex is misleading, and that the distinction it elides is our thesis: **drift is observed; whether encoder adaptation contributed must be intervened upon**. We ask what drift cannot answer—did *allowing* encoder adaptation contribute to the gain?—with a **frozen-encoder intervention** (B–D) behind a downstream-only **pretrained-capability inclusion criterion**. **Our headline finding is methodological: B–D must be scored on *held-out* windows**. Read on the windows model selection already uses—the natural protocol, and the one we previously reported—it systematically flatters the frozen encoder. Re-scored on a disjoint test split, three of the **19 cells** we previously reported **reverse sign** (e.g. Chronos/ETTh1  $+6.8 \rightarrow -39.2$  pp), the criterion reclassifies two others, and a third backbone added here reverses a fourth—all four the same way. Across 23 cells B–D spans  $-39$  to  $+62$  pp. Second, across four TSFMs (Moirai, Chronos, TimesFM, MOMENT), drift *magnitude* (CKA) does not reliably determine the *direction* of that effect—not even after backbone identity is absorbed as a fixed effect rather than by subsetting (App. F). Third, the intervention exposes a reproducible regime **within Moirai** that we name **pretrained-capability degradation**: in six cells meeting the criterion ( $R_{\text{task}}^2$  0.50–0.83) full fine-tuning leaves the model **worse than not fine-tuning at all** ( $+8.7\%$  to  $+46.1\%$  above zero-shot, in every seed) while a frozen encoder *improves* it ( $-3.5\%$  to  $-16.0\%$ ); a criterion fixed before any test window is consulted selects the same six. Those six are among the **least-drifted** we measure (CKA 0.67–0.89), while the cells adaptation helps most sit at 0.09–0.63: at this resolution the drift reflex would have pointed the wrong way. We establish this regime’s **existence, not its prevalence**—the third backbone contains no such cell. Two analyses from an earlier version are withdrawn and support no claim here (§4).

## 1 Introduction

A Moirai encoder fine-tuned on ETTh2 can have CKA with its pretrained self fall to  $\sim 0.3$ —a  $\sim 70\%$  representational shift—and yet forecast *better* on the very same task. If drift were a proxy for damage, this should not happen.

Practitioners who fine-tune released TSFM checkpoints inherit a reflex from continual learning: measure drift, read large drift as catastrophic forgetting, and reach for preservation methods (LoRA, EWC, L2-SP) [Kirkpatrick et al., 2017, Hu et al., 2022, Li et al., 2018]. This bundles three implicit claims—drift implies forgetting, forgetting implies harm, drift should be minimized—and the bundle comes apart on TSFMs. Our “ $\neq$ ” is insufficiency, not independence: drift does not *reliably imply* harm, because **downstream utility depends on the functional role of the changed representation**—not *how much* a backbone drifts but *what kind* of change it is, which one scalar cannot distinguish.

We contribute a within-model dissociation (Fig. 2), a capability check over four architecturally distinct TSFMs with controlled encoder interventions on the cells that pass it, and a diagnostic chain needing only released weights and target data (Fig. 1). On scope: the check spans four backbones, the 23 intervention cells come from Moirai, Chronos and TimesFM, and MOMENT supports no claim here. Three hypotheses, separated throughout. **H1** does drift predict utility? **H2** does drift predict how much *allowing* encoder adaptation contributes? **H3** does fine-tuning actually destroy a demonstrated pretrained capability? We give evidence against reliable monotonicity for H1 and H2; H3 yields **pretrained-capability degradation**, a reproducible regime *within Moirai*. Ordering these needs one prior result—*where* the intervention is scored—which is why that comes first below.

Our claims, strongest evidence first. (1) **B–D must be scored out-of-sample**—read on the windows model selection uses, it flatters the frozen encoder, four cells reverse sign, and the gate licensing them is vulnerable the same way. (2) **CKA drift does not reliably determine the direction of the encoder-adaptation contribution** under our evaluated protocols. (3) **A downstream-only inclusion criterion** marks where pretrained capability is demonstrated; it is not a validated detector, and we show what it would have selected prospectively. (4) **The reading is predictable without the frozen control** within Moirai, from condition B’s own held-out forgetting—a partly algebraic rule that does not transfer (§4).

## 2 Setup and the Diagnostic Framework

We study four TSFMs: **Moirai** [Woo et al., 2024] (encoder-only, MoE), **Chronos-T5** [Ansari et al., 2024] (encoder-decoder), **TimesFM** [Das et al., 2024] (decoder-only,  $d_{\text{model}}=1280$ ), and **MOMENT** [Goswami et al., 2024] (encoder-only). Data is univariate ETT [Zhou et al., 2021]/M4 (lookback 96;  $h=24$  cross-backbone,  $h=96/192$  Moirai sweep), plus ILI (weekly influenza, 7 features, lookback 104,  $h=24$ ). **Each arm trains through its own objective and head**. The 14 Moirai cells—which include all six degradation cells—and ILI train with Moirai’s *own* NLL objective through its *own* distribution head (`param_proj`) and are scored through its native sampling path (median of 20 samples); no regression head is attached, and the same holds for the 4 TimesFM cells, which train and are scored through TimesFM’s *own* point/quantile head (App. E). Only the 5 Chronos cells attach a common MSE head to encoder states, because its generative decoder admits no matched frozen-versus-adapted comparison otherwise—hence limitation (iii). What departs from the released recipes is the *data* pipeline (one dataset at a time, fixed lookback, no LOTSA packing), not the objective, head or inference path; the cost is that MSE magnitudes are *not* comparable across arms, so B–D too is read within a backbone. Two conditions run throughout: **B** (full fine-tuning) and **D** (frozen encoder, head-only). Forgetting% =  $100(\text{MSE}_{\text{FT}} - \text{MSE}_{\text{ZS}})/\text{MSE}_{\text{ZS}}$ ; negative means improvement over zero-shot.

Three quantities must not be conflated: **representation drift** (CKA), **adaptation contribution** (B–D) and **downstream utility** (forecast MSE)—CKA measures the first, B–D the second, neither is the third. One inclusion criterion, one detector, one intervention:

**Pretrained-capability inclusion criterion.**  $R_{\text{task}}^2 = 1 - \text{MSE}_{\text{ZS}}/\text{MSE}_{\text{Linear}}$ : does the released model beat a lookback-96 linear baseline by  $>20\%$ ? It asks whether there is capability worth preserving—not whether fine-tuning is worthwhile—and needs only released weights and the practitioner’s series. **Its role is inclusion, not prescription: it decides which cells make an intervention interpretable, and we do not claim it is a validated deployment-time decision rule**—it identifies cells with measurable zero-shot capability; we do not show that consulting it improves fine-tuning choices. A gate-fail means no demonstrated zero-shot advantage *under this protocol*, not that the pretrained representation carries no transferable information.

**CKA drift.** Linear CKA [Kornblith et al., 2019] between pretrained and fine-tuned backbone on mean-pooled encoder outputs over identical held-out 192-timestep windows; flagged at  $< 0.95$  and compared only *within* a backbone, never across architectures. Crucially, **CKA measures representational change, not functional change**, which is why the chain does not stop here. We are not arguing that CKA is a poor similarity measure: the claim is against *equating* its magnitude with downstream damage, not against the measurement. Both cutoffs are *operational detectors within this protocol*, not semantic thresholds: pooling moves the CKA magnitude but not the flag (App. M, I).

**Frozen-encoder intervention (B–D).** Condition D measures how much encoder adaptation *contributes* beyond head-only training; B–D is the resulting forgetting gap in percentage points, negative

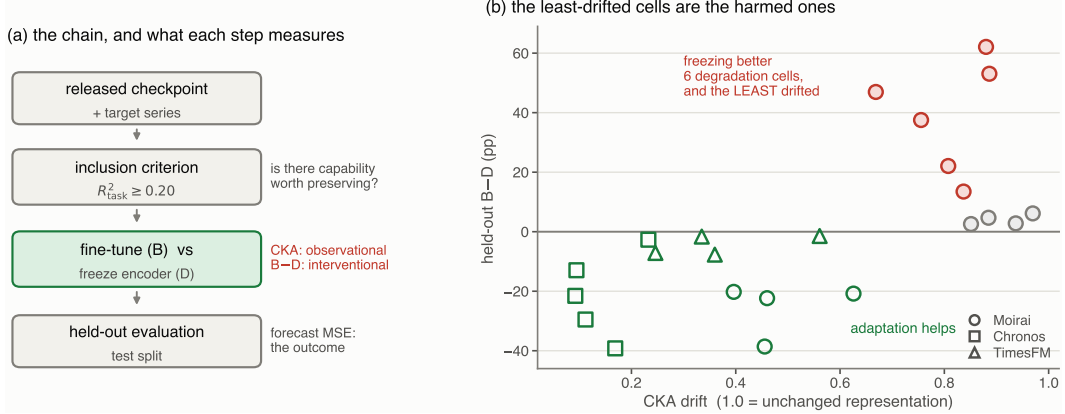


Figure 1: **The chain, and why drift cannot stand in for the intervention.** (a) The three quantities the chain keeps apart. (b) All 23 cells; red marks the six **pretrained-capability degradation** cells. They are the *least*-drifted cells we measure (CKA 0.67–0.89) while adaptation helps most at 0.09–0.46, so the reflex—large drift means damage—points the wrong way. The  $x$ -axis is not a decision boundary: CKA is comparable only *within* a backbone, and within Moirai it does not reliably order the column ( $\rho=+0.495$ , CI includes 0; App. F).

Table 1: **The intervention spans 101 pp; drift does not order it.** Held-out B-D (pp): negative = adaptation helps, **positive = freezing is better**. 23 cells, three backbones, 3–10 seeds; gate and B-D both scored on the test split. <sup>†</sup>gate-fails there despite passing on the selection windows; \*directional ( $|\text{mean}| < 2 \text{ SEM}$ ); <sup>‡</sup>B-D only—Chronos’s head/decoder mismatch makes per-condition forgetting non-comparable (limitation iii). **degradation** = pretrained-capability degradation (§3); Fig. 1b plots all 23. Per-cell values App. G, statistics in App. F.

| Backbone             | Data ( $h$ ), gate $R^2_{\text{task}}$        | CKA       | B-D held-out          | Reading            |
|----------------------|-----------------------------------------------|-----------|-----------------------|--------------------|
| Chronos <sup>‡</sup> | ETTh1 / ETTm2 / Weather (24), .39–.65         | 0.09–0.17 | −39.2 / −29.5 / −21.5 | adapt              |
| Chronos <sup>‡</sup> | ETTh2 .46 / Electricity .14 <sup>f</sup> (24) | 0.09–0.23 | −13.0 / −2.7*         | adapt              |
| Moirai-S             | ILI (24), −.03 <sup>f</sup>                   | 0.455     | −38.6                 | adapt              |
| Moirai-B             | ETTh2 (96 / 192), .29 / .55                   | 0.40–0.46 | −22.3 / −20.2         | adapt              |
| Moirai-L             | ETTh2 (96), .13 <sup>f</sup>                  | 0.626     | −20.8                 | adapt              |
| TimesFM              | ETTh2 / ETTm2 (24), .70 / .47                 | 0.25–0.36 | −7.7 / −7.1           | adapt              |
| TimesFM              | Weather .51 / ETTh1 .40 (24)                  | 0.34–0.56 | −1.6* / −1.4*         | directional        |
| Moirai-S             | ETTh2, ETTm2 (96 / 192), .33–.72              | 0.85–0.97 | +2.6 to +6.2*         | directional        |
| Moirai-S             | ETTh1 (96 / 192), .58 / .72                   | 0.81–0.84 | <b>+22.1 / +13.5</b>  | <b>degradation</b> |
| Moirai-B             | ETTh1 (96 / 192), .50 / .68                   | 0.67–0.76 | <b>+37.5 / +47.0</b>  | <b>degradation</b> |
| Moirai-S             | Weather (96 / 192), .53 / .83                 | 0.88–0.89 | <b>+53.1 / +62.1</b>  | <b>degradation</b> |

93 meaning adaptation helps, and it is the only step that manipulates the encoder rather than observing  
94 it. Freezing applies to encoder weights only—input/output projections stay trainable—so D is not  
95 representation-frozen (its CKA runs 0.38–1.000). **B-D therefore measures whether encoder**  
96 **adaptation was necessary for the gain—not that the representation change CKA measures**  
97 **carried it.** Freezing also removes optimization degrees of freedom and gradient pathways, so no  
98 single mechanism is isolated by this contrast.

### 99 3 Results: What the Intervention Separates

100 Table 1 summarizes the cross-backbone results, ordered by B-D. One structural fact first: **the**  
101 **criterion is horizon-sensitive**. At  $h=96$  Chronos, TimesFM and MOMENT all gate-fail (−14% to  
102 −148%); re-screened at  $h=24$  on held-out windows, both backbones we re-screen flip to gate-pass—  
103 Chronos on four of five datasets, TimesFM on all five ( $R^2_{\text{task}}$  +0.21 to +0.70), tracking Chronos  
104 dataset for dataset. So pretrained value is a property of checkpoint  $\times$  dataset  $\times$  horizon, not of an  
105 architecture; MOMENT is not re-screened and carries no claim here (App. E).

106 **Detection: where B-D is scored decides its sign.** B-D compares two trained models and inherits  
107 their model selection: B early-stops on the evaluation windows and a frozen head’s regularizer is

108 tuned there, so scoring B–D on those windows—the natural protocol, and the one we previously  
109 reported—is no generalization estimate. We re-scored all 19 cells on a chronologically disjoint test  
110 split, fitting and selecting no model on it (App. G; dispersions are  $\pm 1$  SEM unless a table states *std*).  
111 Three cells *reverse sign*—Chronos/ETTh1  $+6.8 \pm 2.9 \rightarrow -39.2 \pm 13.9$ , and both Moirai/ETTh2 cells  
112  $(+3.8, +7.2 \rightarrow -22.3, -20.8)$ —while others merely soften. The error runs one way: read on its  
113 selection windows, B–D *flatters the frozen encoder*. TimesFM’s four cells, scored on both splits  
114 from the same runs and so never re-scored, reverse a fourth the same way  $(+0.2 \rightarrow -7.1)$ : four of 23,  
115 all one direction. **And so does the gate—which makes cell inclusion retrospective.** Recomputed  
116 there, two cells that pass on the selection windows fail out of sample (Chronos/Electricity 0.14,  
117 Moirai-Large/ETTh2 0.13; ILI fails on both sides, App. C), and the values are no longer bimodal, so  
118 the 20% threshold is not margin-free (App. M). Applying the criterion to cells whose test split we have  
119 seen makes the six below *retrospectively* identified, so we re-ran the definition behind a **prospective**  
120 gate from validation windows alone: it is the more permissive of the two (22 of 23 cells, not 20)  
121 and selects the **same six** (App. G). **Intervention: pretrained-capability degradation (H3).** Our  
122 headline empirical finding is a reproducible regime *within Moirai* in which full encoder fine-tuning  
123 *destroys* the pretrained advantage the criterion demonstrated while a frozen-encoder control preserves  
124 it. Three clauses must hold, the last two in *every* seed: the cell gate-passes, B ends up worse than  
125 the un-tuned model, and D improves on it. Six cells qualify. Moirai-Small/Weather is clearest—it  
126 beats a lookback-96 linear baseline by 53%/80% at  $h=96/192$ , yet B lands  $+37.3\%/+46.1\%$  *above*  
127 zero-shot error while D reaches  $-15.8\%/-16.0\%$ —and Moirai-Base and Moirai-Small repeat it on  
128 ETTh1 at smaller magnitude (B  $+8.7\%$  to  $+39.5\%$  vs D  $-3.5\%$  to  $-14.7\%$ , gates 0.50–0.72), so  
129 the regime spans two model sizes (App. G).

130 **Robustness.** Freezing the input projections too (condition H, CKA exactly 1.0 by construction) keeps  
131 the sign on all six cells and moves the magnitude by at most 5.8 pp (App. H)—“freezing wins” is  
132 not the input projection re-fitting. It does split the second clause: **6/6** cells beat zero-shot under D,  
133 **5/6** under the stricter H (Moirai-S/ETTh1  $h=192$ ,  $\text{forg}_H=+1.2$ ). Chronos/M4 (gate 84.5%) anchors  
134 the stable end (App. L). **A third backbone.** TimesFM-2.5 under its *own* recipe—no head attached,  
135 so both forgetting terms are clean—gate-passes on four datasets, and **none is a degradation cell**:  
136 B improves on zero-shot in all four, B–D runs  $-7.7$  to  $-1.4$ . The regime is Moirai’s, at horizons  
137 where TimesFM gate-fails: scope, not generality. It also gives the measurement claim its sharpest  
138 form: condition D’s transformer stack is **bit-identical** after training ( $\ell_2$  drift exactly 0.00) and still  
139 reads CKA 0.38–0.89—all of it the input tokenizer—so low CKA cannot by itself mean a damaged  
140 encoder (App. E).

141 **Scope: drift does not flag the degradation cells.** The six sit at CKA 0.67–0.89 while the cells  
142 adaptation helps most (Chronos/ETTh1  $-39.2$ , ILI  $-38.6$ ) sit at 0.17 and 0.46—the reflex would  
143 have cleared the damaging cells and flagged the safe ones. Nor does it order the held-out column  
144 *within* a backbone— $\rho=+0.495$  over 14 Moirai cells, and the slope stays unpinning when backbone  
145 identity is absorbed as a fixed effect instead ( $n=23$ , App. F). Within one model they come apart  
146 directly (Fig. 2, App. A).

## 147 4 Discussion

148 **What this is and is not.** Two claims, and only one is prospective: the sign reversals are a generaliza-  
149 tion result, since no model is fitted or selected on test, while the degradation regime is *identified* using  
150 test data and so establishes existence, not prevalence or detection accuracy (App. G). Runs were not  
151 blinded, thresholds were set after inspecting results, and two pre-specified predictions were falsified.

152 **A cheaper proxy, where it holds.**  $B-D \equiv \text{forg}_B - \text{forg}_D$  and Moirai’s frozen arm barely moves, so  
153 condition B’s own forgetting agrees in sign with B–D in 12 of 14 cells ( $\rho=+0.956$ )—one run, no  
154 control, but partly algebraic and it does not transfer (App. J).

155 **What follows for practice.** Check that the checkpoint beats a trivial baseline on your series before  
156 worrying about preserving it; do not read drift as damage, because CKA is an observation, not a  
157 verdict; and if you compare a frozen encoder against an adapted one, score it on *held-out* windows—  
158 on selection windows the comparison favours freezing, and four of our cells reverse.

159 **Limitations.** Related work: App. K. (i) Two analyses from an earlier version are **withdrawn**—the  
160 probe axis, and our largest reported effect, whose frozen control had never been trained; B–D shares  
161 no machinery with either (App. B, I, C). (ii) Cells are unequally powered (3–10 seeds) and **not**

162 **independent**—datasets, horizons and sizes recur—so effective  $n$  is far below 23 and cross-cell  
 163 statistics are descriptive; directional cells are not findings. (iii) Chronos carries no per-condition  
 164 forgetting (§ in Table 1). (iv) The six degradation cells are Moirai-only on two datasets: existence,  
 165 not prevalence. (v) Condition D leaves input projections trainable (App. H bounds that cost). (vi)  
 166 Every data pipeline departs from the released recipe (§2).

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## A Within-Model Dissociation Sweep

Every point shares a backbone, a dataset, a horizon and a protocol; only the fine-tuning sample size varies. Increasing  $n$  moves drift monotonically while the task outcome moves non-monotonically: CKA falls steadily from 0.949 at  $n=500$  to 0.518 at  $n=10k$ , while forgetting goes  $+14.0$ ,  $+13.4$ ,  $-2.9$ ,  $+6.6$ ,  $-5.3\%$ . The best cell in the sweep is the *most-drifted* one. The  $n=5k$  variance ( $+6.6\%$ , std 10.0, 7 seeds) is bimodal—4 seeds near  $-0.1\%$ , 3 at  $+15.5\%$  with an epoch-1 overshoot—so we treat that point as unresolved rather than as evidence for either direction.

This is a *dissociation*, not an *intervention on drift*. Sample size is an uncontrolled causal variable: raising it changes supervision, optimization trajectory and overfitting, not drift alone. The sweep therefore shows that drift and outcome need not move together within one model; it cannot show that drift itself is causally inert. The frozen-encoder experiment is the intervention, and it is what the paper’s claims rest on.

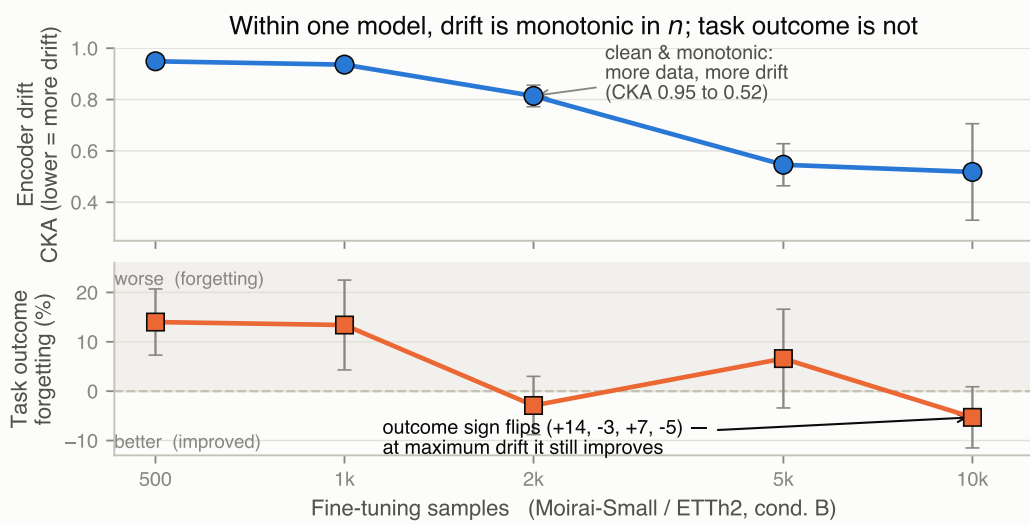


Figure 2: **Within-model dissociation.** Moirai-Small/ETTh2 ( $h=96$ , cond. B) across  $n$ : as  $n$  rises CKA drift is monotonic (top) while forgetting is non-monotonic and sign-flipping (bottom), improving at maximum drift: **drift and downstream outcome can move independently**. Sample size is uncontrolled here, so this demonstrates **non-monotonic co-movement, not causal independence**: it is a dissociation, not an intervention on drift. Error bars  $\pm 1$  SD;  $n=10k$  is 10-seed CUDA-deterministic.

## B The Probe Evidence We Retract

**This appendix records a retracted result.** It reports the probe statistics as we originally computed them, unchanged, so that they can be audited against the re-analysis in Appendix I that withdraws them. Nothing in the main results depends on this section; we keep it rather than deleting the numbers.

Table 2 gives every probe used in the asymmetry test on Moirai-Small/ETTh2, with dispersion and seed counts. We classified each  $\Delta R^2$  by an effect-size heuristic—whether the mean exceeds one seed-level SD—not by a significance test. Four auxiliary probes decrease; one is indistinguishable from zero; none increase. The signature as stated—trained increases **and** no auxiliary increases—is satisfied without relying on the forecast-48 result in either direction. What it is *not* robust to is the operating point, which is the subject of Appendix I.

**Why these five.** The auxiliary set is chosen to vary along two axes a single probe family would confound: *what* is decoded (low-order statistics vs. forecasts) and *how* it is decoded (linear Ridge vs. gradient-boosted). The delta1 probe matters specifically because its Ridge version is misleading: the zero-shot Ridge baseline on next-step deltas is deeply negative ( $R^2 \approx -2.3$ ), so any encoder shift inflates  $\Delta R^2$  (Ridge gives  $+1.04 \pm 0.22$ ). A GBM probe establishes a genuine positive floor

Table 2: Probe  $\Delta R^2$  on Moirai-Small/ETTh2,  $h=96$ ,  $n=10k$ . Auxiliary targets span statistical properties (mean, variance, lag-1 autocorrelation), an alternative forecast horizon (forecast-48), and a positive-floor readout (GBM delta1).

| Probe target                 | Trained? | Seeds | $\Delta R^2$                   | Classification           |
|------------------------------|----------|-------|--------------------------------|--------------------------|
| Forecast-96 (objective)      | yes      | 10    | $+0.67 \pm 0.23$ (10/10 pos)   | <b>increase</b>          |
| Mean                         | no       | 10    | $-0.033 \pm 0.019$ (10/10 neg) | decrease                 |
| Variance                     | no       | 10    | $-0.048 \pm 0.030$ (9/10 neg)  | decrease                 |
| Lag-1 autocorrelation        | no       | 10    | $-0.052 \pm 0.041$ (9/10 neg)  | decrease                 |
| Delta1 (GBM, positive floor) | no       | 4     | $-0.056 \pm 0.030$             | decrease                 |
| Forecast-48 (sub-horizon)    | no       | 4     | $-0.001 \pm 0.163$             | <b>indist. from zero</b> |

( $R^2(\text{PT})=+0.059$ ) and on that floor decodability *falls*. We report the GBM result for this reason, and note that this is the only positive-floor untrained readout available to us—a reviewer-visible weakness rather than a strength. Five probes cannot establish that no specialization occurs along an unmeasured axis (calibration, periodicity, tail behaviour); the asymmetry test is evidence, not proof.

## C ILI Cell Detail

**The gate value we previously reported for this cell does not reproduce.** Earlier versions quoted gate-pass =57% (ZS MSE 1.183, Linear MSE 2.748) from a run set that was not retained. Re-computed from the runs this paper actually reports (10 seeds, results/v40\_ili\_heldout) with the same definition and the same per-window normalisation, the gate is  $+0.139$  on validation (ZS 1.893, linear 2.206) and  $-0.033$  on the test segment (ZS 2.479, linear 2.664): **ILI gate-fails on held-out windows**, and we withdraw the 57%. The per-feature picture is why the mean lands where it does—five of seven features are positive but only two clear 0.20, and NUM. OF PROVIDERS is  $-0.86$  because that series is close to a straight line on the test segment, which linear extrapolation fits almost exactly. ILI is not a degradation cell and no claim in §3 rests on its gate; what changes is that it can no longer be cited as a moderate-gate cell, and the “falsified prediction” below is correspondingly weaker evidence—a heavily restructuring encoder on a cell with *no* demonstrated zero-shot advantage is much less surprising than one on a moderate-gate cell.

**Status of this cell.** On ILI the encoder restructures heavily (CKA  $0.455 \pm 0.008$ ) and full fine-tuning improves by  $-80.0 \pm 0.6\%$  (10/10), falsifying a prediction specified beforehand (0.90–0.95 CKA expected). We previously reported B–D= $-80.7$  pp here, i.e. that encoder adaptation supplied *all* of the gain. That is withdrawn. The frozen control it was measured against had never been trained: the script assigned the zero-shot MSE as the fine-tuned MSE and wrote a forgetting of exactly 0.0, so the “0.0% in all 10 seeds” was one constant, not ten measurements. Re-run so that condition D actually trains — encoder frozen, the 3.2M parameters outside it updated — the gap is  $-62.6 \pm 1.4$  on validation and  $-38.6 \pm 1.3$  held-out (10/10 negative on both). D reaches CKA 0.76–0.90, not 1.000, because the projections feeding the encoder remain trainable. Condition B is unaffected and reproduces the published  $-80.7 \pm 0.9$  and CKA 0.478 to within seed noise, so the entire discrepancy was the untrained control. Note also that both references are weak in absolute terms (ZS MSE 1.893, linear 2.206 on validation).

The cell is Moirai-Small on national influenza-like illness: 966 weekly observations, 7 features,  $h=24$ , lookback 104, split 579/193/194 chronologically. Condition D, re-run so that it trains, gives CKA  $0.827 \pm 0.013$  and forgetting  $-17.4 \pm 1.3\%$  on validation ( $-12.2 \pm 1.3\%$  held-out), 10/10 negative: head-only training recovers a substantial share of the improvement, so the B–D gap is  $-62.6$  pp on validation and  $-38.6$  pp held-out, not the  $-80.7$  previously reported.

**Per-feature asymmetry.** The forecast target OT shows the largest gain ( $+0.73 \pm 0.16$ , 10/10 positive) and AGE 0–4 also gains consistently ( $+0.15 \pm 0.05$ , 10/10). AGE 5–24 is negative ( $-0.07 \pm 0.09$ , 3/10 positive). Decodability thus improves on the target and degrades on a series the head was not optimizing—the same signature as the ETTh2 trained-vs-auxiliary split, obtained here within a single cell.

**Falsified prediction.** We specified CKA 0.90–0.95 for ILI in advance, on the expectation that a moderate gate implies moderate drift. The observed 0.456 falsifies this: gate strength does not linearly

Table 3: ILI per-seed results (Moirai-Small,  $h=24$ , lookback 104), both conditions, regenerated for this revision. Each value is the mean across the 7 univariate features. The  $\Delta R^2$  column of the earlier version of this table is dropped with the retracted probe axis; condition D is shown because it previously did not exist as a measurement.

| Seed | B (full fine-tuning) |                         | D (frozen encoder) |                         |
|------|----------------------|-------------------------|--------------------|-------------------------|
|      | CKA                  | Forg.% (val / held-out) | CKA                | Forg.% (val / held-out) |
| 42   | 0.458                | −78.6 / −51.5           | 0.804              | −16.2 / −14.6           |
| 101  | 0.465                | −81.4 / −50.1           | 0.898              | −13.5 / −6.7            |
| 123  | 0.476                | −80.0 / −48.9           | 0.756              | −27.6 / −19.5           |
| 202  | 0.457                | −81.4 / −52.1           | 0.791              | −19.7 / −13.6           |
| 303  | 0.409                | −78.1 / −49.0           | 0.823              | −18.3 / −11.5           |
| 456  | 0.463                | −75.5 / −50.0           | 0.851              | −15.4 / −13.2           |
| 777  | 0.437                | −81.6 / −52.5           | 0.822              | −18.0 / −12.4           |
| 789  | 0.449                | −80.3 / −49.5           | 0.865              | −12.7 / −4.9            |
| 888  | 0.505                | −82.0 / −50.8           | 0.848              | −16.1 / −12.8           |
| 999  | 0.437                | −80.8 / −53.4           | 0.807              | −16.1 / −12.7           |
| Mean | 0.456                | −80.0 / −50.8           | 0.827              | −17.4 / −12.2           |

predict drift magnitude, and the restructuring-to-stability transition sits nearer  $\approx 80\%$  gate than the 50–70% we assumed. We report this because it is the kind of prediction the chain is supposed to expose.

## D Full Preservation Spectrum

**What this appendix is for.** It is a *secondary*, observational result: it shows that raising representational similarity does not raise utility, which is evidence against the reflex’s prescriptive half. It is **not** evidence for this paper’s causal claim—that rests entirely on the frozen-encoder intervention B–D in §3, because these methods preserve representations by different mechanisms and so are not a controlled manipulation of one variable. The main text quotes  $h=96$  only. Table 4 gives both horizons and all methods, including the NLL+SupCon variant. These methods preserve representations by *different mechanisms*—penalty (L2-SP, EWC), reparameterization (LoRA), and complete freezing—so CKA is a common observational axis, not a common intervention, and at  $n=4$ –7 the correlations here are descriptive only. We stress that **LoRA does well**: the paper’s claim is that drift magnitude fails to *order* these methods, not that preservation is a bad idea or that fine-tuning should be avoided. **A correction to an earlier version.** The figures we previously reported for this spectrum were not matched to their baseline. The mitigation runs use  $n=1000$  while the full-fine-tuning reference we quoted against them was an  $n=500$ , 10-seed run, and the mitigation forgetting percentages divided by a single shared zero-shot (0.1259) rather than each run’s own, which sits between 0.1274 and 0.1310; that inflated every one of them by 1.2–3.4 pp. Corrected—all at  $n=1000$ ,  $h=96$ , each run against its own zero-shot—full fine-tuning is CKA .936 at +13.4%, against L2-SP  $\lambda=0.01$  (.939, +16.0), L2-SP  $\lambda=0.1$  (.965, +13.4), EWC  $\lambda=1000$  (.972, +14.9) and LoRA (.979, −9.2).

The claim this displaces is the strong one: it is *not* true that every weight-anchoring configuration forecasts worse. L2-SP  $\lambda=0.1$  raises CKA substantially (.936  $\rightarrow$  .965) and lands on full fine-tuning to within 0.02 pp. What the corrected spectrum shows is the weaker and more useful thing: moving up the CKA axis buys nothing predictable—one anchoring setting is worse, one is a dead heat, a third is worse again, and the reparameterizing method is far better while being the most CKA-preserving of the four. Representation preservation does not predict utility here. At  $h=192$  no  $n=1000$  full-fine-tuning baseline exists in our runs, so we report that horizon’s methods without a matched reference rather than against the  $n=500$  one.

**Why we do not report a single CKA–utility correlation.** A natural request is a correlation over Table 4 rather than a verbal reading of seven points. We computed it, and the answer is instructive: *the sign depends on which mechanisms are pooled.*



Table 4: Mitigation spectrum on ETTh2 forecasting. Methods ordered by representation preservation (CKA,  $h=96$ ). LoRA preserves representations (CKA $\approx 0.98$ ) while *improving* over zero-shot. Mean  $\pm$  std; Forgetting% = (fine-tuned MSE – zero-shot MSE) / zero-shot MSE  $\times$  100. **Two caveats on comparability, added in revision.** Forgetting% divides by the single shared zero-shot printed in each horizon header, not by each run’s own zero-shot; the anchoring and LoRA runs measure their own at 0.127–0.131, so their figures here are inflated by 1.2–3.4 pp. And the rows are not matched on training-set size: B and C use  $n=500$  training samples while L2-SP, EWC and LoRA use  $n=1000$  (the  $n$  column counts seeds, not samples). Matched at  $n=1000$ ,  $h=96$ , each against its own zero-shot: full fine-tuning .936/+13.4, L2-SP  $\lambda=0.01$  .939/+16.0, L2-SP  $\lambda=0.1$  .965/+13.4, EWC .972/+14.9, LoRA .979/–9.2. On that comparison L2-SP  $\lambda=0.1$  ties full fine-tuning rather than losing to it, so anchoring is not uniformly harmful; the dissociation the table is cited for—CKA not predicting utility—is unaffected.

| Method                         | $h = 96$ (ZS MSE: 0.126) |             |                         |                       | $h = 192$ (ZS MSE: 0.158) |             |                         |                       | $n$ |
|--------------------------------|--------------------------|-------------|-------------------------|-----------------------|---------------------------|-------------|-------------------------|-----------------------|-----|
|                                | MSE                      | Forg.%      | CKA                     | Drift                 | MSE                       | Forg.%      | CKA                     | Drift                 |     |
| B: Full fine-tune              | .140 $\pm$ .014          | +11.1       | .937 $\pm$ .028         | 3.96 $\pm$ .56        | .183 $\pm$ .020           | +15.7       | .970 $\pm$ .009         | 4.12 $\pm$ .32        | 10  |
| F: L2-SP ( $\lambda=0.01$ )    | .150 $\pm$ .006          | +19.0       | .939 $\pm$ .023         | 4.98 $\pm$ .34        | .179 $\pm$ .009           | +13.3       | .945 $\pm$ .007         | 4.84 $\pm$ .22        | 5   |
| C: NLL+SupCon                  | .138 $\pm$ .010          | +10.1       | .949 $\pm$ .014         | 3.72 $\pm$ .43        | .178 $\pm$ .011           | +12.9       | .964 $\pm$ .015         | 3.97 $\pm$ .34        | 10  |
| F: L2-SP ( $\lambda=0.1$ )     | .147 $\pm$ .003          | +16.7       | .965 $\pm$ .006         | 2.94 $\pm$ .08        | .181 $\pm$ .003           | +14.6       | .980 $\pm$ .002         | 2.84 $\pm$ .02        | 5   |
| G: EWC ( $\lambda=1000$ )      | .149 $\pm$ .015          | +18.3       | .972 $\pm$ .004         | 3.29 $\pm$ .79        | .188 $\pm$ .004           | +19.0       | .978 $\pm$ .006         | 4.46 $\pm$ .38        | 5   |
| E: LoRA ( $r=8$ ) <sup>†</sup> | <b>.116</b> $\pm$ .001   | <b>–7.9</b> | .979 $\pm$ .006         | <b>0.00</b> $\pm$ .00 | <b>.157</b> $\pm$ .002    | <b>–0.6</b> | .986 $\pm$ .005         | <b>0.00</b> $\pm$ .00 | 5   |
| D: Frozen encoder              | .126 $\pm$ .006          | +0.5        | <b>1.000</b> $\pm$ .000 | 1.11 $\pm$ .15        | .155 $\pm$ .002           | –1.7        | <b>1.000</b> $\pm$ .000 | 1.30 $\pm$ .18        | 10  |

<sup>†</sup> LoRA ( $\alpha=16$ ) adds low-rank updates that do not modify base encoder parameters;  $\ell_2$  drift is exactly zero by construction.

| Conditions included             | $h=96$      |                 | $h=192$     |                 |
|---------------------------------|-------------|-----------------|-------------|-----------------|
|                                 | Pearson $r$ | Spearman $\rho$ | Pearson $r$ | Spearman $\rho$ |
| Weight-anchoring only ( $n=4$ ) | +0.46       | +0.40           | +0.64       | +0.40           |
| All seven conditions ( $n=7$ )  | –0.58       | –0.54           | –0.60       | –0.43           |

Forgetting is negative-is-better, so a *positive* coefficient means higher CKA accompanies worse utility. Among weight-anchoring methods the sign is positive at both horizons; pooling all seven flips it negative, because LoRA and the frozen encoder sit at the highest CKA and perform well. Neither coefficient is meaningful on its own at  $n=4$  or  $n=7$ , and we claim no significance. The point is the flip: CKA is a common *observational* axis across these methods, not a common intervention, so no single correlation summarizes the table. This is the same reason we present the spectrum as a mechanism-separated comparison rather than a regression.

## E Chronos and TimesFM Intervention Detail

**Chronos-T5-Small, moderate gate ( $n=8k$ , conditions B and D).** This pair is the main text’s first inversion. An earlier version of this table reported it from runs that gave both conditions the same 30-epoch budget. That budget is generous for B, which early-stops with its best epoch at 1–2, but it truncates D badly: every D seed restored “best epoch 30 of 30” with validation loss still falling nearly linearly, so the apparent gap measured the budget rather than the intervention. We re-ran both cells with each condition trained to its own convergence.

Because in condition D the encoder is frozen *and* in `eval()` mode, its mean-pooled output is a deterministic function of the context window, and the head is a single linear map on that vector. Caching the pooled features is therefore exact, not an approximation, and it makes convergence affordable. We then estimate the best head two independent ways: a closed-form ridge optimum over 16 log-spaced penalties (the headline, since a closed form admits no epoch budget at all; the penalty is selected on the same eval windows B early-stops on, so both sides receive eval-set selection, and the optimum was interior to the grid in all six runs), and a protocol-matched AdamW run to early stopping. The two agree to about 1 pp everywhere, which is the evidence that the number is a property of the frozen features rather than of the optimizer.

One further consequence of the re-run is why Table 1 reports B–D for these cells but no per-condition forgetting. Forgetting is a ratio against the zero-shot reference, and for Chronos the fine-tuned

Table 5: Chronos-T5-Small,  $h=24$ ,  $n=8k$ , matched seeds and eval windows. MSE is per-element validation MSE;  $B-D=(MSE_B - MSE_D)/MSE_{ZS} \times 100$ , so the zero-shot reference cancels and only the two same-head models are compared. All six D runs converged (early stopping fired) and all six ridge optima were interior.

| Cell           | Seed | MSE <sub>B</sub> | D (ridge optimum) |          | B-D (pp)        |                 |         |
|----------------|------|------------------|-------------------|----------|-----------------|-----------------|---------|
|                |      |                  | MSE <sub>D</sub>  | $\alpha$ | ridge           | AdamW           | CKA (B) |
| ETTh1          | 42   | 1.598            | 1.580             | 0.3      | +1.2            | +0.5            | 0.179   |
|                | 43   | 1.471            | 1.364             | 0.03     | +7.5            | +5.9            | 0.158   |
|                | 44   | 1.509            | 1.365             | 0.1      | +11.4           | +11.1           | 0.171   |
| mean $\pm$ std |      |                  |                   |          | +6.7 $\pm$ 5.1  | +5.8 $\pm$ 5.3  | 0.169   |
| ETTh2          | 42   | 0.998            | 1.133             | 0.03     | -13.3           | -14.8           | 0.094   |
|                | 43   | 0.842            | 1.053             | 0.03     | -23.4           | -25.0           | 0.102   |
|                | 44   | 0.882            | 1.078             | 0.03     | -22.8           | -21.7           | 0.087   |
| mean $\pm$ std |      |                  |                   |          | -19.9 $\pm$ 5.7 | -20.5 $\pm$ 5.2 | 0.094   |

model is scored through the trained MSE head while that reference is scored through Chronos’s generative decoder—so the ratio charges the output-head swap to forgetting and is not a measurement of adaptation at all. B-D is immune because both conditions share the head and the reference cancels algebraically. An earlier version of the table did quote those per-condition numbers, and the direction of the effect they implied does not survive: measured condition B is +12.6% on ETTh1 and -1.6% on ETTh2. It also quoted a Chronos CKA range of 0.886–0.949, which we have traced to the Moirai-Small sample-size sweep; the measured Chronos values are 0.087–0.179.

On the validation windows every ETTh1 seed sits above every ETTh2 seed, which an earlier version read as an opposite-signed pair. That separation does not survive held-out scoring: on the test windows both cells are negative (ETTh1 -39.1 $\pm$ 24.1, ETTh2 -12.7 $\pm$ 1.0, 3/3 each), so we no longer report a sign contrast between them (App. G). What the two cells still show is that CKA does not order the outcome—ETTh1 drifts *less* (0.158–0.179 against 0.087–0.102) and loses *more* to freezing.

**Chronos-T5-Small, strong gate.** On M4-Monthly (gate 84.5%, 10 seeds) condition B gives +0.5% and condition D +0.6% (10 seeds; std 2.7 and 5.3); the B-D gap is within the frozen-control noise floor, and a sign test on the 6/10 positive-forgetting seeds is not distinguishable from chance ( $p=0.75$ , two-sided binomial).

**Cross-dataset transfer.** Fine-tuning Moirai-Small on ETTh2 improves ETTh1 (-7.7 $\pm$ 2.5%) and ETTm2 (-3.3 $\pm$ 1.0%), 3 seeds, condition B; the frozen control shows near-zero out-of-distribution effect.

**Pretraining overlap.** Moirai’s LOTSA corpus excludes ETT; M4 is in Chronos’s pretraining data (which is why the M4 cell functions as an overlap control); overlap for TimesFM and MOMENT could not be established from public documentation.

**TimesFM-2.5-200M at  $h=96$ .** Five seeds on ETTh1 across conditions B and D give stability throughout (CKA 0.995–0.998,  $\Delta R^2 \approx 0$ , no asymmetry), and the checkpoint gate-fails at this horizon ( $R^2_{\text{task}}$  -0.139 on ETTh1, -0.138 on ETTh2), so our own criterion does not license an intervention there. Per-seed forgetting figures for the two conditions were not retained, so this pair is not one of the 19 re-scored cells of Table 1 rather than being entered with an estimate.

**TimesFM-2.5-200M at  $h=24$ : the criterion.** Because the criterion is horizon-sensitive, we re-screened TimesFM at  $h=24$  before running any intervention, on the *same* held-out window set the Chronos arm uses—univariate OT, lookback 96, the 200-window seed-0 test subsample—so the criterion, the zero-shot reference and B-D all read one window set. All five datasets pass, and they track Chronos dataset for dataset, which is evidence that  $R^2_{\text{task}}$  is measuring a property of the dataset  $\times$  horizon rather than of the backbone:

| $R^2_{\text{task}}$ at $h=24$ (test) | ETM2   | Weather | ETTh2  | ETTh1  | Electricity |
|--------------------------------------|--------|---------|--------|--------|-------------|
| TimesFM-2.5-200M                     | +0.703 | +0.506  | +0.472 | +0.401 | +0.214      |
| Chronos-T5-Small                     | +0.651 | +0.494  | +0.456 | +0.389 | +0.138      |

Both arms are scored by `scripts/gate_all_cells.py`, one code path for all four arms, cached to `results/gate_test_side.json`.

**TimesFM-2.5-200M at  $h=24$ : the intervention.** Having licensed it, we run the same conditions as the Moirai arm (`scripts/finetune_timesfm.py`): **B** trains everything; **D** freezes the 20-layer stack (196.71M of 231.29M parameters) and leaves the input tokenizer and output projections trainable, exactly as Moirai’s **D** leaves `in_proj` trainable. Hyperparameters are the ones the other arms use— $\text{lr } 10^{-4}$ , AdamW, grad-norm clip 1.0, 20 epochs, patience 7, 1000 training windows—and are identical for **B** and **D**, since **B**–**D** would be meaningless otherwise. After restoring the best epoch we *assert* the frozen modules are bit-identical to the released checkpoint; the run aborts rather than mislabel a condition.

Two design points matter for reading these cells. First, no head is attached: the loss is TimesFM’s own output structure (MSE on the mean channel plus pinball loss on the nine decile channels the released output `projection_point` emits), and zero-shot, **B** and **D** are all scored through the same native path. **Per-condition forgetting is therefore a clean measurement here**, unlike Chronos (limitation iii), so this arm re-tests the `forB` rule on a second backbone. Second, the released torch API cannot be trained through—`module.decode()` is wrapped in `torch.no_grad()` and returns detached NumPy—so training uses a differentiable re-implementation of the prefill path. That path is legitimate only because at  $h=24$  the autoregressive loop runs  $(\text{max\_horizon} - 1)/o = 0$  times with  $o=128$ : one forward pass is the whole forecast. `verify_native_path()` asserts agreement with `model.forecast()` on real windows before any training (max relative deviation  $6.5 \times 10^{-7}$  on MPS, 0.0 on CPU), so a re-implementation error cannot reach the paper silently. The continuous quantile head (27.85M) rewrites only non-median channels at inference and receives no gradient, so condition **B** updates 203.44M of the 231.29M parameters.

**What the four cells show.** We ran the four highest-gate datasets, 3 seeds  $\times$  {**B**, **D**} each, in an order fixed before any run by correspondence to the six degradation cells (ETTh1 first, then Weather, then ETM2, ETTh2). **All four land in the *adapt* half of Table 1, and none is a degradation cell:** condition **B** improves on zero-shot everywhere (`forB`  $-51.8\%$  to  $-3.0\%$ ), which is the first clause of degradation failing, so the regime does not extend to this backbone at  $h=24$ . That is a scope limit on H3 (limitation iv), not a refutation of it: the six degradation cells sit at  $h=96/192$ , where TimesFM *gate-fails* and our own criterion forbids the comparison.

| TimesFM, $h=24$ | gate   | CKA (B)   | CKA (D)   | <code>for<sub>B</sub></code> | <code>for<sub>D</sub></code> | <b>B</b> – <b>D</b> val | <b>B</b> – <b>D</b> held-out     |
|-----------------|--------|-----------|-----------|------------------------------|------------------------------|-------------------------|----------------------------------|
| ETM2            | +0.703 | 0.19–0.52 | 0.48–0.55 | $-51.8 \pm 2.8$              | $-44.1 \pm 1.6$              | $-12.2 \pm 12.4$        | <b><math>-7.7 \pm 2.6</math></b> |
| Weather         | +0.506 | 0.24–0.40 | 0.62–0.71 | $-7.8 \pm 1.6$               | $-6.1 \pm 0.5$               | $-4.8 \pm 0.6$          | <b><math>-1.6 \pm 1.5</math></b> |
| ETTh2           | +0.472 | 0.22–0.28 | 0.38–0.50 | $-20.4 \pm 1.4$              | $-13.3 \pm 1.9$              | $+0.2 \pm 2.9$          | <b><math>-7.1 \pm 3.1</math></b> |
| ETTh1           | +0.401 | 0.44–0.64 | 0.80–0.89 | $-3.0 \pm 2.5$               | $-1.6 \pm 0.9$               | $-5.4 \pm 0.9$          | <b><math>-1.4 \pm 2.8</math></b> |

Three things the arm does establish, none of which was available from Moirai or Chronos.

**(a) The held-out reversal replicates on fresh cells.** These four cells were scored on both splits from the same runs—nothing was re-scored after the fact—and TimesFM/ETTh2 *reverses sign*,  $+0.2$  on validation against  $-7.1$  held out. It reverses in the *same* direction as the three Moirai/Chronos reversals: validation favoured the frozen encoder, held-out data does not. Four of our 23 cells reverse and all four reverse that way, which is the paper’s central methodological claim (§3) reproduced on a backbone it was not derived from.

**(b) A stack that is provably unchanged can read CKA 0.38.** Condition **D**’s  $\ell_2$  weight drift over `stacked_xf` is exactly 0.00—the bit-identity assertion above—yet its measured CKA runs 0.38–0.89. Every point of that movement comes from the input tokenizer changing what the frozen stack is shown. This is the sharpest form of the paper’s measurement claim: CKA registers representational change on a network whose weights did not move at all, so a low CKA cannot by itself be read as the encoder having been damaged. It also bounds limitation (v) from the other side—**D** is not representation-frozen anywhere, and here we can say exactly why.

(c) **Drift does not order the four.** CKA (B) spans 0.19–0.64, comparable to the degradation cells’ drift, while held-out B–D stays within 8 pp of zero; the ranking is not monotone ( $\rho = +0.400$ ,  $n=4$ , CI  $[-0.778, +1.000]$ ), and the *most*-drifted cell (ETTh2, CKA 0.22–0.28) is among the two where adaptation helps most. Because no head is attached,  $\text{forg}_B$  is a clean measurement here, so this arm also re-tests the §4 rule: its *sign* transfers— $\text{forg}_B < 0$  and  $B-D < 0$  in 4 of 4 cells—but its stated mechanism does not, since  $\text{forg}_D$  varies here almost as much as  $\text{forg}_B$  (SD 19.2 against 22.0, versus 7.5 against 31.6 on Moirai). The rule is an empirical property of a nearly-constant frozen arm, and that property is Moirai’s, not a general one.

We do not re-screen MOMENT: its  $h=96$  margin is  $-147.8\%$ , and its released checkpoint has no pretrained forecasting head, so the zero-shot term the criterion needs does not exist without fitting a probe—which would make the “zero-shot advantage” an artifact of the probe. MOMENT therefore supports no claim in this paper.

## F Cross-Cell Statistics, and a Pooled Analysis We Do Not License

Across the 23 cells of Table 1 we ask whether either observational diagnostic orders the held-out intervention. Spearman  $\rho$  with a 10,000-sample bootstrap CI over cells:

| Grouping                       | $n$ | CKA range | $\rho$ | 95% CI           |
|--------------------------------|-----|-----------|--------|------------------|
| Pooled across backbones        | 23  | 0.09–0.97 | +0.741 | [+0.511, +0.838] |
| Moirai only                    | 14  | 0.40–0.97 | +0.495 | [−0.153, +0.808] |
| Chronos only                   | 5   | 0.09–0.23 | +0.100 | [−1.000, +1.000] |
| TimesFM only                   | 4   | 0.25–0.56 | +0.400 | [−0.778, +1.000] |
| $\ell_2$ weight drift (pooled) | 17  | —         | −0.301 | [−0.731, +0.249] |

The  $\ell_2$  row covers 17 cells rather than 23 because the Chronos runs did not retain a weight-drift measurement. **A correction to an earlier version:** that row previously read  $n=19$ ,  $\rho = +0.044$ , which was in fact the Moirai-only figure ( $n=13$ , CI  $[-0.545, +0.651]$ ) carrying a “pooled” label. The pooled value is  $-0.301$ ; both include zero, so the reading does not change, but the label was wrong and the sign was not ours to report.

**Absorbing the backbone instead of subsetting it.** The within-backbone rule can be read as convenient—we adopt the split that removes the significant relationship. So we also fit the pooled cells directly, with backbone identity absorbed as a fixed effect rather than by subsetting, and CKA entering as a continuous regressor (`scripts/cka_fixed_effects.py`):

| Model for held-out B–D             | $n$ | $b_1$ (pp per unit CKA) | 95% CI           |
|------------------------------------|-----|-------------------------|------------------|
| $b_0 + b_1$ CKA (no backbone term) | 23  | +60.5                   | —                |
| $b_0 + b_1$ CKA + backbone FE      | 23  | +87.1                   | [−48.8, +168.8]  |
| + CKA $\times$ backbone            | 23  | +95.5                   | [−215.8, +224.6] |

CI’s are a 20,000-draw bootstrap resampling *clusters*, a cluster being a (backbone, dataset) pair, because datasets and checkpoints recur across horizons and sizes and an unclustered interval would be far too narrow; draws that leave a backbone unrepresented are discarded and the retained count is printed by the script. **Both adjusted intervals include zero.** The pooled rank correlation therefore does not survive absorbing backbone identity, and it is not the within-backbone restriction that removes it—the same conclusion follows from a model fitted on all 23 cells at once. What the data cannot do is pin the within-backbone slope down; that is weaker than, and must not be read as, a claim that CKA carries no information.

**The pooled row is not a licensed analysis, and it points the other way.** Pooled, CKA orders the intervention with a CI excluding zero ( $p=0.0001$ ). We do not read it as evidence, for the reason stated in §2: CKA magnitude is representation-dependent, so values from different architectures are not on a common scale, and we compare them only within a backbone. That restriction was fixed before these cells were run, not chosen after seeing this table. The pooled figure is largely what an architecture confound would produce here: every Chronos cell sits at CKA 0.09–0.23 and every Moirai cell at 0.40–0.97, so pooling those two recovers the backbone split rather than a drift–utility

relationship. The third backbone is the honest complication: TimesFM’s four cells span 0.25–0.56 and therefore *straddle* the Chronos and Moirai ranges instead of forming a third stratum, so the pooled correlation is not purely a between-backbone artifact. It is also the group where drift is largest relative to outcome—four cells at CKA 0.25–0.56 whose B–D all sit within 8 pp of zero—so adding it pushes the pooled  $\rho$  up while adding cells that individually contradict the reflex. Neither observation licenses reading the pooled row as evidence; both are why we print it.

We report it anyway because a reader can compute it in a minute, and because the honest position is that our central claim survives on a protocol restriction rather than because no correlation can be found in the numbers. Within Moirai, the only group with enough cells and enough CKA range to test,  $\rho = +0.495$  with a CI that includes zero—but at  $p=0.072$  this is marginal, and we do not claim CKA is uninformative, only that it does not reliably order the intervention at this resolution. Within TimesFM, the one other backbone with a clean forgetting measurement,  $\rho$  is  $+0.400$  on 4 cells with a CI spanning almost the whole range—too few cells to test anything, and reported so that the arm’s failure to help is on the record rather than omitted. The  $\ell_2$  drift result is cleaner: it orders nothing even pooled ( $\rho = -0.301$ , CI  $[-0.731, +0.249]$ ), which is the more robust form of the claim that observational summaries of fine-tuning do not substitute for running the intervention.

## G Held-Out Re-Scoring of the Intervention

**Why this appendix exists.** Every B–D we previously reported was read on the evaluation windows that model selection already used: condition B early-stops on them, and for the converged Chronos runs the frozen head’s ridge penalty is chosen on them. That makes B–D a controlled within-cell contrast but not an estimate of generalization. Each training script in this work already carves a chronologically disjoint test split and then discards it; here we score it.

**Two kinds of evidence, and only one of them is prospective.** These are separate claims and the paper does not conflate them. (A) *The protocol finding is genuinely out of sample.* No model is fitted or selected on the test split, so the sign reversals in §3 are a generalization result. (B) *The degradation regime is a retrospective discovery.* Its definition includes a gate that we *recompute* on the test split, so the six cells are identified using data we have already looked at. That establishes existence; it is not a prospective validation of the criterion, and it says nothing about how often the regime occurs or how well the criterion would detect it in advance.

**What a prospective gate would have selected.** Because (B) is the weaker claim, we measure it rather than merely caveat it. `scripts/gate_all_cells.py -split val` recomputes  $R_{\text{task}}^2$  from validation windows alone—each arm’s own stored validation zero-shot MSE over that arm’s own validation windows, with no test data touched anywhere—and `cell_matrix.py -gate-split=val` then re-applies the full definition with the forgetting terms left test-side, so only the *inclusion* decision changes. The prospective gate is the *more permissive* of the two, admitting 22 of 23 cells against the test-side gate’s 20 (the two it additionally admits, Chronos/Electricity 0.30 and Moirai-Large/ETTh2 0.60, both read *adapt*; ILI fails on both sides at  $+0.139$  and  $-0.033$ ). It selects the **same six degradation cells**, at validation gates 0.32–0.62 against 0.50–0.83 on test. The regime is therefore not an artifact of gating on the split it is measured on—though it remains a retrospective characterisation in every other respect, and we do not claim the criterion is a validated detector.

**A correction to the gate itself.** Auditing the above surfaced a window mismatch in the Moirai arm of `gate_all_cells.py`: the linear denominator was built with an extended lookback of  $2 \times 96$  for every cell, while `finetune_forecasting.py` evaluates with lookback  $+h$ . The two agree at  $h=96$  and diverge at  $h=192$ , where the numerator and denominator of  $R_{\text{task}}^2$  were being averaged over different target windows. Corrected, six  $h=192$  gate values move (largest, Moirai-Base/ETTh2  $0.72 \rightarrow 0.55$ ; Moirai-Small/Weather  $0.80 \rightarrow 0.83$ ). **No cell changes gate status and the degradation set is unchanged**; the values in Table 1, Table 6 and App. M are the corrected ones.

**Protocol.** Train on train, select on validation, report on test. Nothing is selected on the test windows: condition B’s early stopping is untouched, and the ridge penalty is still the argmin of the *validation* grid, merely re-scored at that  $\alpha$  on test. The Chronos runs share one fixed test subsample across seeds and one seeded zero-shot reference per dataset, since B–D divides by a shared denominator. Condition B was re-trained because only the encoder, not the head, had been checkpointed; its validation losses reproduce the published values to 0.008–0.039% (MPS run-to-run float noise), and all six frozen-encoder ridge optima reproduce exactly and remain interior.

510 **Which cells count as degradation.** The definition—gate-passing,  $\text{forg}_B > 0$  in every seed,  $\text{forg}_D < 0$   
511 in every seed—selects exactly six of the 23 cells, and `cell_matrix.py` applies it programmatically  
512 so the list cannot be curated by hand. The four TimesFM cells are put through the same test and none  
513 passes it. One further cell (Moirai-Small/ETTh2,  $h=192$ ) meets the inequalities on *means* but its  
514 harm term is only +2.0 with 5/10 seeds positive, so it is excluded and reported as directional. An  
515 earlier version of the paper listed four, labelling the two Moirai-Small/ETTh1 cells *preserve* by hand  
516 despite their satisfying every clause.

517 Table 6 is emitted by `scripts/cell_matrix.py`, the same function that computes the body’s  
518 numbers, rather than transcribed. An earlier version of this appendix listed four cells while Table 1  
519 already quoted held-out values for all 19—the appendix had not been updated when the re-scoring  
520 was extended, and the mismatch reads as the body overclaiming. Generating it is the only fix that  
521 stays fixed. All dispersions here and in the body are SEM; the earlier table mixed SEM with SD,  
522 which made two cells look better separated than they were.

Table 6: Every cell, both scorings. Gate and B–D are computed on the same test windows; <sup>f</sup> marks the three cells that gate-fail there. “Reverses” = the sign of B–D differs between the two scorings.

| Cell                                    | Seeds | Gate $R_{\text{task}}^2$ | B–D validation | B–D held-out       | Reverses? |
|-----------------------------------------|-------|--------------------------|----------------|--------------------|-----------|
| Chronos / ETTh1 ( $h=24$ )              | 3     | +0.389                   | +6.8±2.9       | – <b>39.2±13.9</b> | yes       |
| Moirai-S / ILI ( $h=24$ )               | 10    | –0.033 <sup>f</sup>      | –62.6±1.4      | – <b>38.6±1.3</b>  | no        |
| Chronos / ETTh2 ( $h=24$ )              | 3     | +0.651                   | –24.1±9.8      | – <b>29.5±6.4</b>  | no        |
| Moirai-B / ETTh2 ( $h=96$ , $n=1000$ )  | 3     | +0.287                   | +3.8±4.1       | – <b>22.3±2.9</b>  | yes       |
| Chronos / Weather ( $h=24$ )            | 3     | +0.494                   | –22.4±4.5      | – <b>21.6±3.2</b>  | no        |
| Moirai-L / ETTh2 ( $h=96$ , $n=500$ )   | 3     | +0.129 <sup>f</sup>      | +7.2±5.8       | – <b>20.8±1.4</b>  | yes       |
| Moirai-B / ETTh2 ( $h=192$ , $n=1000$ ) | 3     | +0.554                   | –3.9±6.2       | – <b>20.2±1.1</b>  | no        |
| Chronos / ETTh2 ( $h=24$ )              | 3     | +0.456                   | –19.9±3.2      | – <b>12.9±0.6</b>  | no        |
| TimesFM / ETTh2 ( $h=24$ )              | 3     | +0.703                   | –12.2±12.4     | – <b>7.7±2.6</b>   | no        |
| TimesFM / ETTh2 ( $h=24$ )              | 3     | +0.472                   | +0.2±2.9       | – <b>7.1±3.1</b>   | yes       |
| Chronos / Electricity ( $h=24$ )        | 3     | +0.138 <sup>f</sup>      | –10.3±3.2      | – <b>2.7±1.9</b>   | no        |
| TimesFM / Weather ( $h=24$ )            | 3     | +0.506                   | –4.8±0.6       | – <b>1.6±1.5</b>   | no        |
| TimesFM / ETTh1 ( $h=24$ )              | 3     | +0.401                   | –5.4±0.9       | – <b>1.4±2.8</b>   | no        |
| Moirai-S / ETTh2 ( $h=96$ , $n=1000$ )  | 5     | +0.568                   | +6.5±2.9       | + <b>2.6±3.4</b>   | no        |
| Moirai-S / ETTh2 ( $h=96$ , $n=500$ )   | 10    | +0.331                   | +10.6±2.8      | + <b>2.8±3.7</b>   | no        |
| Moirai-S / ETTh2 ( $h=192$ , $n=1000$ ) | 5     | +0.722                   | +8.6±3.3       | + <b>4.7±3.2</b>   | no        |
| Moirai-S / ETTh2 ( $h=192$ , $n=500$ )  | 10    | +0.670                   | +17.4±4.3      | + <b>6.1±3.4</b>   | no        |
| Moirai-S / ETTh1 ( $h=192$ , $n=1000$ ) | 5     | +0.719                   | +14.8±1.9      | + <b>13.5±1.7</b>  | no        |
| Moirai-S / ETTh1 ( $h=96$ , $n=1000$ )  | 5     | +0.578                   | +16.1±1.5      | + <b>22.1±2.3</b>  | no        |
| Moirai-B / ETTh1 ( $h=96$ )             | 3     | +0.505                   | +15.8±4.6      | + <b>37.5±8.2</b>  | no        |
| Moirai-B / ETTh1 ( $h=192$ )            | 3     | +0.682                   | +15.6±2.3      | + <b>47.0±13.8</b> | no        |
| Moirai-S / Weather ( $h=96$ )           | 3     | +0.526                   | +55.4±4.3      | + <b>53.1±10.2</b> | no        |
| Moirai-S / Weather ( $h=192$ )          | 3     | +0.825                   | +70.2±12.3     | + <b>62.1±13.6</b> | no        |

523 **What moves.** Condition B is stable across the two segments; condition D is not. On Chronos/ETTh1  
524 the frozen head goes from 1.58/1.36/1.37 on validation to 1.81/2.51/2.07 on test, while B moves by  
525 a few percent. Chronos/ETTh2, the one cell that does not reverse, is also the one whose frozen head  
526 transfers cleanly (1.133 → 1.135). So the cell where freezing genuinely cost was robust to the choice  
527 of windows, and the cells where freezing appeared to *help* were the ones that did not survive it.

528 **A consequence for the closed-form argument.** We previously justified the ridge head as a closed-  
529 form optimum that admits no epoch budget. It is an optimum *on the validation windows*. Out  
530 of sample a weak validation-selected  $\alpha$  transfers poorly, and the protocol-matched early-stopped  
531 AdamW head—which agrees with ridge to  $\sim 1$  pp on validation—beats it on test (Chronos/ETTh1  
532 seed 43: 2.505 against 1.957). We pre-committed to reporting the estimator validation preferred,  
533 which is ridge in every seed, with AdamW as a sensitivity; taking the better of the two on test would  
534 reintroduce exactly the fault this appendix removes. Under the AdamW leg the ETTh1 reversal is  
535 smaller but intact.

536 **ILI.** This cell also required its condition D to be re-run from scratch, for a different reason: the frozen  
537 control had never been trained (App. C). With a real D its validation B–D is  $-62.6 \pm 1.4$  rather than

the  $-80.7$  we reported, and  $-38.6 \pm 1.3$  on held-out windows. Its D reaches CKA 0.76–0.90, not the 1.000 the earlier version asserted, because the projections feeding the encoder stay trainable.

**Scope and denominators.** Every cell in Table 6 is scored on a chronologically disjoint test split. The zero-shot test denominator matters more than it looks: B–D divides by a shared zero-shot term, and each run stores one measured on *validation*, so using it against a test-window difference mixes scales—on Moirai-Small/ETTh2  $h=96$  the test zero-shot is 0.492 against validation’s 0.129, which inflates held-out B–D by  $\approx 3.8\times$ . The test references come from three run sets (`results/v41_zs_test`, `v43_moirai_matrix/*/condition_A`, `v39_moirai_zs_test`), 13 in total, and the Chronos cells share one fixed test subsample and one seeded zero-shot per dataset across seeds. Two rows of Table 1 in an earlier version—Chronos/M4 and the TimesFM  $h=96$  pair—have no paired B/D artifacts with test MSEs retained and are not among the 23; they appear only in App. E. The four TimesFM  $h=24$  cells are a different matter: they were run for this revision and scored on test windows from the outset, so nothing about them was re-scored.

## H The Strict-Freeze Control

Condition D freezes the encoder’s own weights but leaves the input projection (`in_proj`, 247,680 parameters) and `mask_encoding` (384) trainable alongside the distribution head (`param_proj`, 2,956,800). What the encoder *receives* therefore keeps changing, so its output is not a fixed function of the input—which is why D’s measured CKA is 0.76–0.90 on ILI and 0.97–0.99 on Moirai-Base rather than 1.000. A reader can reasonably object that “freezing wins” in the degradation cells is partly the input projection re-fitting rather than the encoder being held still.

Condition H removes the objection: `in_proj` and `mask_encoding` are frozen too, so only `param_proj` trains and the encoder’s output is pinned by construction (measured CKA 1.0000 exactly, as it must be). B–H is then a clean read of what allowing *any* encoder-side change was worth, and B–D versus B–H separates encoder adaptation from input re-fitting. We fixed the reading rule before running: the claim is that the sign and reading of B–D survive strict freezing, so B–H is reported beside B–D on every cell, and divergence on any cell is reported as that cell’s gap being partly input re-fitting rather than being buried.

**Result.** All six degradation cells and ILI were run; nothing was dropped. Measured CKA is 1.0000 everywhere, confirming the construction. **The reading survives on every cell:** B–H keeps B–D’s sign in all seven, and the magnitude changes by  $-5.8$  to  $+0.5$  pp—so at most 5.8 pp of a gap as large as 62 pp is input-projection re-fitting. On ILI, where adaptation helps, strict freezing makes adaptation look *more* valuable ( $-38.6 \rightarrow -43.6$ ), which is the same direction: holding more of the model still costs more where the encoder was earning its keep. B–D is averaged over the seeds condition H ran, so Moirai-Small/ETTh1’s entries here are 3-seed values against the 5-seed values quoted in Table 1.

| Cell                                    | Seeds | B–D              | B–H                                | <code>forgh</code> | CKA <sub>H</sub> |
|-----------------------------------------|-------|------------------|------------------------------------|--------------------|------------------|
| Moirai-S / Weather ( $h=192$ )          | 3     | $+62.1 \pm 13.6$ | <b><math>+56.7 \pm 13.5</math></b> | $-10.6 \pm 0.3$    | 1.0000           |
| Moirai-S / Weather ( $h=96$ )           | 3     | $+53.1 \pm 10.2$ | <b><math>+50.1 \pm 6.6</math></b>  | $-12.8 \pm 0.6$    | 1.0000           |
| Moirai-B / ETTh1 ( $h=192$ )            | 3     | $+47.0 \pm 13.8$ | <b><math>+47.5 \pm 13.7</math></b> | $-8.0 \pm 0.4$     | 1.0000           |
| Moirai-B / ETTh1 ( $h=96$ )             | 3     | $+37.5 \pm 8.2$  | <b><math>+32.5 \pm 8.2</math></b>  | $-9.7 \pm 0.4$     | 1.0000           |
| Moirai-S / ETTh1 ( $h=96$ , $n=1000$ )  | 3     | $+23.2 \pm 3.6$  | <b><math>+20.4 \pm 3.6</math></b>  | $-0.6 \pm 0.2$     | 1.0000           |
| Moirai-S / ETTh1 ( $h=192$ , $n=1000$ ) | 3     | $+14.5 \pm 1.4$  | <b><math>+8.7 \pm 1.5</math></b>   | $+1.2 \pm 0.4$     | 1.0000           |
| Moirai-S / ILI ( $h=24$ )               | 10    | $-38.6 \pm 1.3$  | <b><math>-43.6 \pm 0.7</math></b>  | $-7.1 \pm 0.5$     | 1.0000           |

**One clause does not survive, and it is not the headline one.** “Degradation” has two parts: fine-tuning ends up worse than the un-tuned model, *and* freezing improves on it. The first part is a property of condition B and is untouched by this control. The second is condition D’s, and on Moirai-Small/ETTh1  $h=192$  the strictly-frozen model does *not* beat the un-tuned model: `forgh` =  $+1.2 \pm 0.4\%$ , positive in all 3 seeds, where `forgh` is  $-4.6 \pm 0.6\%$ . On that cell the head alone cannot recover the zero-shot forecast; D’s improvement came partly from re-fitting the input projection. The other five cells keep both parts (`forgh` from  $-0.6\%$  to  $-12.8\%$ ), and B–H stays positive on all six, so “freezing beats full fine-tuning” holds throughout while “freezing also beats not fine-tuning” holds strictly on five of six. We report the cell rather than re-defining the term around it.

## I CKA Pooling Robustness, and the Re-Analysis that Retracts the Probe Axis

A consequence worth stating plainly: because the magnitude of CKA depends on how the representation is constructed, this paper’s claims are strongest where they are *ordinal and interventional*—which cells drift, and what freezing costs on each—and weakest if read as statements about absolute CKA values. The drift *flag* is stable under pooling; the number attached to it is not.

**Pooling robustness.** CKA is reported on mean-pooled encoder outputs throughout. Because mean-pooling discards token-level structure, we recomputed whole-encoder CKA on the same ten fine-tuned ETTh2 encoders ( $n=10k$ ,  $h=96$ , condition B) both mean-pooled and token-flattened, over 500 held-out 192-timestep windows.

| Representation            | CKA (mean $\pm$ SD, 10 seeds) | Seed range  |
|---------------------------|-------------------------------|-------------|
| Mean-pooled (as reported) | 0.561 $\pm$ 0.157             | 0.328–0.840 |
| Token-flattened           | 0.732 $\pm$ 0.036             | 0.685–0.797 |

Two things follow. The *qualitative* drift flag is robust: both values sit far below the 0.95 threshold, so every cell keeps its assignment. The *magnitude* is not: token-level similarity is substantially higher and far less seed-variable ( $\pm 0.036$  vs  $\pm 0.157$ ), so part of the drift reported as  $\sim 0.52$  is an artifact of pooling. We regard this as consistent with, rather than damaging to, the thesis—it is a further reason not to read a single scalar as a measure of damage—but CKA magnitudes should be read as pooling-dependent.

**Probe protocol, recovered and verified.** The probe pipeline was not documented in enough detail to re-run, so we reconstructed it from source and verified it against the published numbers. The protocol is: the encoder input window is **192 steps** (extended lookback = lookback + horizon), the probe **trains on the first 300 validation windows and is evaluated on the first 200 test windows**, the target is the next 96 steps across **all seven features** flattened, and the probe is Ridge( $\alpha=1.0$ ) on mean-pooled encoder outputs. Under exactly this protocol the trained probe reproduces:

| Trained 96-step probe | $R^2$ (PT) | $R^2$ (FT) | $\Delta R^2$ (10 seeds)        |
|-----------------------|------------|------------|--------------------------------|
| Published             | −6.90      | −6.23      | +0.67 $\pm$ 0.23, 0/10 neg     |
| This reproduction     | −6.9043    | −6.2364    | +0.6679 $\pm$ 0.2262, 0/10 neg |

The headline  $\Delta R^2$  therefore reproduces to four significant figures from the released checkpoints. Three protocol details each move the result by an order of magnitude and are easy to get wrong: using the 192-step window rather than a 96-step context, evaluating across the validation→test boundary rather than within one split, and using all seven features as the target rather than the target channel alone. Substituting a within-validation split, for instance, moves  $R^2$ (PT) from −6.90 to −0.13.

**A transparent re-analysis, and what it costs us.** Rather than keep searching for the target normalization behind the published auxiliary floors, we specified a protocol with every knob fixed in advance and report what it gives (scripts/probe\_transparent.py). Windows: the same 500. Features: mean-pooled encoder output, standardized on probe-train. Probe: Ridge with  $\alpha$  chosen by 5-fold CV on the *pretrained* encoder’s probe-train over  $\log_{10} \alpha \in [-3, 6]$ , then that same  $\alpha$  reused for all ten fine-tuned encoders so the two are never tuned apart. Statistic:  $\Delta R^2$  per seed with a percentile bootstrap 95% CI over the ten seeds (10,000 resamples). Two additions decide the outcome:

- **Two splits.**  $T$  is the published temporal split (train on validation, test on test).  $R$  randomizes the same 500 windows into 300/200. The pair separates decodability from the validation→test level shift.
- **A permuted-target noise floor.** The identical pipeline run on 20 independently shuffled copies of each target. Any real  $\Delta R^2$  must exceed the spread the pipeline produces when the target carries no information at all.

**The published  $\Delta R^2$  is a regularization artifact.** Holding the split at  $T$  and sweeping  $\alpha$ , the trained probe’s  $\Delta R^2$  decays monotonically—+0.668, +0.271, +0.066, +0.034, +0.005, +0.000 at  $\alpha = 1, 10, 10^2, 10^3, 10^4, 10^5$ —while  $R^2$ (PT) stays between −6.90 and −5.23 throughout. The effect is largest exactly where the probe extrapolates worst, and CV on the training split selects



Table 7: Transparent protocol, Moirai-Small/ETTh2, 10 released checkpoints. Under the published split  $T$  no probe generalizes ( $R^2(\text{PT}) < 0$ ) and every  $|\Delta R^2|$  is far below the permuted-target floor. Under  $R$ , where every probe generalizes, no  $|\Delta R^2|$  exceeds 0.011 and the trained probe *decreases*.

| Split | Target                | $R^2(\text{PT})$ | $\Delta R^2$ | 95% CI               | floor  |
|-------|-----------------------|------------------|--------------|----------------------|--------|
| $T$   | Forecast-96 (trained) | -5.23            | -0.0005      | $[-0.001, +0.000]$   | 0.0005 |
|       | Mean                  | -164.09          | +3.22        | $[-3.60, +10.00]$    | 102.1  |
|       | Variance              | -20.50           | -8.22        | $[-11.94, -4.14]$    | 132.3  |
|       | Lag-1 autocorr.       | -2.71            | -0.80        | $[-2.02, +0.16]$     | 30.7   |
| $R$   | Forecast-96 (trained) | +0.678           | -0.0042      | $[-0.0061, -0.0025]$ | 0.019  |
|       | Mean                  | +0.994           | -0.0014      | $[-0.0022, -0.0006]$ | 0.226  |
|       | Variance              | +0.937           | +0.0025      | $[-0.0014, +0.0068]$ | 0.169  |
|       | Lag-1 autocorr.       | +0.750           | +0.0103      | $[+0.0012, +0.0194]$ | 0.223  |

628  $\alpha=10^6$ , at which it vanishes. Under split  $R$ , where the trained probe attains  $R^2(\text{PT}) = +0.68$ ,  $\Delta R^2$   
629 is at most  $+0.017$  at any  $\alpha$  and is negative at the CV-selected value. There is no configuration in  
630 which the trained probe both generalizes and increases by anything near  $+0.67$ . (Standardizing the  
631 targets is provably *not* a knob: with an intercept, Ridge is equivariant under affine target scaling, and  
632 the sweep confirms the rows are identical to the last digit.)

633 **This contradicts our own stated criterion.** Section B rejects the Ridge version of the delta1 probe  
634 precisely because “the zero-shot Ridge baseline is deeply negative ( $R^2 \approx -2.3$ ), so any encoder shift  
635 inflates  $\Delta R^2$ .” The headline trained probe sits at  $R^2(\text{PT}) = -6.90$ —further into that regime than the  
636 probe we disqualified—and we did not apply the criterion to it. Applied consistently, it disqualifies  
637 the published trained/auxiliary comparison as well.

638 **What survives.** Under the one split where all seven probes generalize, every  $|\Delta R^2|$  is  $\leq 0.011$ :  
639 the trained probe decreases ( $-0.0042$ , 10/10 seeds), mean and trend decrease, lag-1 autocorrelation  
640 *increases*, and variance, lag-24 and spectral centroid are indistinguishable from zero. That is not  
641 the claimed asymmetry; it is closer to no detectable change in any direction. We therefore **retract**  
642 **the probe axis** as unsupported by a protocol we are willing to defend, and we have left the original  
643 numbers in Table 2 rather than deleting them so the two can be compared. The frozen-encoder  
644 intervention is unaffected: it shares no machinery with the probes, manipulates the encoder rather  
645 than fitting a readout on it, and is measured in forecast MSE rather than  $R^2$ . It is where the paper’s  
646 weight now rests, and the two inversions it supports (§3) were never probe-dependent.

## 647 J Predicting the Reading Without the Frozen Control

648 An earlier version listed this as open; within Moirai it is not. Condition B’s *own* held-out forgetting  
649 agrees in sign with B–D in 12 of the 14 Moirai cells (both exceptions directional), and the matched  
650 within-checkpoint contrast is sharper: Moirai-Base forgets under B on ETTh1 at both horizons  
651 ( $+22.8\%$ ,  $+39.5\%$ ) and favours freezing, while the *same* checkpoint on ETTh2 improves under B  
652 ( $-31.4\%$ ) and strongly favours adaptation. One fine-tuning run therefore distinguishes regimes CKA  
653 does not ( $\rho=+0.956$  vs CKA’s  $+0.495$ ,  $n=14$ ).

654 We state the coupling rather than bank it.  $B-D \equiv \text{forg}_B - \text{forg}_D$  identically, and  $\text{forg}_D$  barely moves  
655 (SD 7.5 against  $\text{forg}_B$ ’s 31.6), so the rule works because the frozen arm is nearly constant—an  
656 empirical fact about Moirai, not a theorem. It fails on Chronos ( $\rho=-0.100$ , limitation iii), and on  
657 TimesFM, the other arm where  $\text{forg}_B$  is a clean measurement, the sign agrees 4 of 4 while  $\text{forg}_D$   
658 varies nearly as much as  $\text{forg}_B$  (SD 19.2 vs 22.0): the mechanism does not transfer even where the  
659 rule does. It is reported here, not in the body, because it is a fourth result competing with the three  
660 the paper is actually about, and because a rule that holds on one backbone for a reason that does not  
661 generalise is not a contribution we want weighed alongside them.

## 662 K Related Work

663 Catastrophic-forgetting mitigations—EWC [Kirkpatrick et al., 2017], LwF [Li and Hoiem, 2018],  
664 LoRA [Hu et al., 2022], L2-SP [Li et al., 2018]—are designed to limit how far a fine-tuned model

665 moves from its pretrained state, but they are typically applied without first asking whether the  
666 pretrained state is worth preserving on the target task. Our pretrained-capability inclusion criterion  
667 is a downstream-only check for exactly that question, and the preservation spectrum (Appendix D)  
668 shows why it matters: methods that raise representational similarity do not reliably improve utility,  
669 and the ordering reverses with horizon. The reflex itself is well founded where it originated: Luo  
670 et al. [2023] document substantial forgetting in LLMs under continual fine-tuning. Our claim is not  
671 that the phenomenon is absent but that on TSFMs the drift measurement used to detect it does not  
672 order downstream harm.

673 Closest to this work, Wiliński et al. [2025] study TSFM representations directly: they measure layer  
674 self-similarity within and across model sizes, find block-like redundancy that supports informed  
675 pruning, and steer the latent space to inject concepts such as periodicity and trend. Their interventions  
676 act on a *trained model’s activations* to characterize and control what the representation encodes;  
677 ours acts on the *learning process*—freezing the encoder and re-running fine-tuning—to ask whether  
678 representation change was necessary for the downstream gain. The two are complementary, and their  
679 redundancy finding is a plausible mechanism for what we observe: if TSFM representations carry  
680 substantial redundancy, a large measured drift can be reorganization within that slack rather than  
681 loss of task-relevant structure, which is exactly the gain-independent restructuring our intervention  
682 isolates. Neither their analysis nor ours predicts a cell’s B–D without running both conditions.

683 Kumar et al. [2022] show that fine-tuning can distort pretrained features and that linear probing  
684 followed by fine-tuning mitigates it; Neyshabur et al. [2020] study what transfers between pretrained  
685 and fine-tuned networks. Neither provides a pre-fine-tuning screen for when preservation is warranted,  
686 which is the gap this paper addresses. On the measurement side we use linear CKA [Kornblith et al.,  
687 2019]; our pooling analysis (Appendix I) is a reminder that its magnitude is representation-dependent  
688 and should be read as an operational drift detector rather than a measure of damage.

689 Pretraining overlap for the backbones we test is reported in Appendix E.

## 690 L Moirai-Base Frozen-Encoder Run

691 The Moirai-Base/ETTh2 cell is the most heavily drifted cell in the study but originally had no  
692 frozen-encoder control, so nothing was known about what encoder adaptation contributed there. We  
693 ran condition D for this revision under the published condition-B protocol (Moirai-Base, ETTh2,  
694  $h=96$ ,  $n=10k$ ,  $lr\ 10^{-4}$ , early stopping), on the same ten seeds as the published condition B, on MPS.

|     | Seed     | Forgetting% | Final CKA   | Weight drift | Best epoch |
|-----|----------|-------------|-------------|--------------|------------|
|     | 42       | −49.33      | 0.984       | 1.88         | 7          |
|     | 101      | −54.13      | 0.972       | 2.87         | 16         |
|     | 123      | −45.85      | 0.985       | 1.90         | 6          |
|     | 202      | −40.14      | 0.983       | 1.77         | 6          |
|     | 303      | −43.27      | 0.990       | 1.44         | 4          |
| 695 | 456      | −40.71      | 0.979       | 2.58         | 12         |
|     | 777      | −38.03      | 0.977       | 2.15         | 9          |
|     | 789      | −49.86      | 0.983       | 2.13         | 8          |
|     | 888      | −41.25      | 0.969       | 2.53         | 10         |
|     | 999      | −50.71      | 0.976       | 2.50         | 11         |
|     | Mean±std | −45.33±5.43 | 0.980±0.006 |              |            |

696 All 10 seeds are negative (SEM 1.72). Against the published condition B ( $-54.2\pm 8.6\%$ , 10 seeds)  
697 this is a B–D gap of 8.9 pp: on this cell encoder adaptation supplies a modest fraction of the gain,  
698 despite CKA falling to 0.25–0.33 under full fine-tuning—the deepest drift anywhere in the study. The  
699 first three seeds we ran (42, 101, 123) averaged  $-49.8\%$  and gave a 4.4 pp gap, reported in an earlier  
700 version; extending to the full ten seeds moved D up to  $-45.3\%$  and roughly doubled the gap, so that  
701 earlier estimate rested on a favourable subset. The remaining caveat is hardware: the published B ran  
702 on CUDA and this D on MPS. At ten seeds each the standard deviations still overlap, but the means  
703 no longer do (Welch  $t=2.8$  on the 8.9 pp difference), so this cell is better read as a small-but-real  
704 adaptation contribution than as  $B\approx D$ .

705 **A matched, paired comparison at  $n=1000$ .** A separate Moirai-Base B/D pair exists at  $n=1000$   
706 (20 epochs, no early stopping) with both conditions run on the *same* seeds and hardware, which the  
707  $n=10k$  comparison above cannot offer (its B is the published CUDA run, its D the 10-seed MPS run  
708 reported here).

|     | Seed | B (full FT) | D (frozen) | B−D   |
|-----|------|-------------|------------|-------|
| 709 | 42   | −35.33      | −41.64     | +6.31 |
|     | 123  | −47.93      | −43.62     | −4.31 |
|     | 456  | −47.57      | −56.88     | +9.31 |
|     | Mean | −43.61      | −47.38     | +3.77 |

710 Freezing the encoder is if anything *better* here, in 2 of 3 seeds. Because the seeds and stack match,  
711 this is not exposed to the cross-hardware caveat above, and it reaches the same conclusion at a  
712 different  $n$ : on Moirai-Base, encoder adaptation does not account for the gain.

713 The run also shows that condition D is not representation-frozen. Early stopping selected epochs  
714 6–16 rather than epoch 0, and the frozen-encoder CKA settles at 0.97–0.98 with non-zero weight drift  
715 (1.88–2.87), because the input/output projections outside the encoder remain trainable. In cells where  
716 early stopping selects the untrained state (ILI, Chronos/M4) the measured CKA is 1.000, which is  
717 what earlier drafts described—incorrectly—as holding by construction.

## 718 M Threshold Sensitivity

719 Both thresholds are operational decision boundaries for this study, not universal constants, and neither  
720 was tuned to produce the reported signatures.

721 **Preservation-priority gate (20%), and a margin claim we withdraw.** Earlier versions reported the  
722 observed gate values as bimodal with a wide margin—“no cell falls in  $[0, 28)\%$ , so any threshold in  
723 10–30% produces identical assignments”. That was read on the windows model selection uses. Re-  
724 computed on the test windows for all 23 cells from one code path (`scripts/gate_all_cells.py`,  
725 §3), the values are −3.3, 12.9, 13.8, 28.7, 33.1, 38.9, 40.1, 45.6, 47.2, 49.4, 50.5, 50.6, 52.6, 55.4,  
726 56.8, 57.8, 65.1, 67.0, 68.2, 70.3, 71.9, 72.2, 82.5% (the four TimesFM cells contribute 40.1, 47.2,  
727 50.6 and 70.3). Three cells fall below the threshold (−3.3, 12.9, 13.8) and one sits just above it at  
728 28.7%, so the distribution is not bimodal out of sample and the threshold is *not* margin-free: moving it  
729 to 30% would additionally reclassify Moirai-Base/ETTh2  $h=96$ , and moving it to 10% would restore  
730 Chronos/Electricity and Moirai-Large/ETTh2 to gate-pass. The six degradation cells are unaffected  
731 across the whole 10–50% range (they sit at 50.5–82.5%), which is the assignment the paper’s claims  
732 depend on; the reclassified cells all read *adapt*, where the gate is not what licenses the reading.

733 **CKA threshold reclassification.** The gate analysis above shows wide margins; the CKA cutoff  
734 deserves the same treatment. We reclassify every cell at four thresholds using the observed CKA  
735 values (ranges span seeds, not datasets).

|     | Thresh. | Moirai-B<br>(0.25–0.33) | Moirai/ILI<br>(0.456) | Chronos/ETT<br>(0.087–0.179) | TimesFM ( $h=24$ )<br>(0.19–0.64) | Chronos/M4<br>(0.98–0.999) | TimesFM ( $h=96$ )<br>(0.995–0.999) |
|-----|---------|-------------------------|-----------------------|------------------------------|-----------------------------------|----------------------------|-------------------------------------|
| 736 | 0.900   | drift                   | drift                 | drift                        | drift                             | stable                     | stable                              |
|     | 0.925   | drift                   | drift                 | drift                        | drift                             | stable                     | stable                              |
|     | 0.950   | drift                   | drift                 | drift                        | drift                             | stable                     | stable                              |
|     | 0.975   | drift                   | drift                 | drift                        | drift                             | stable                     | stable                              |

737 Every assignment is identical at all four thresholds, and no cell splits. Note that TimesFM appears  
738 twice and on opposite sides: its  $h=96$  cells are the most stable in the study and its  $h=24$  cells among  
739 the most drifted, so “stable” is not a property of the backbone. The margin is not symmetric, and it  
740 is worth being explicit about which end is tight: the drifting cells sit at least 0.26 below the lowest  
741 cutoff considered, whereas the stable cells clear the highest one by only 0.005 (Chronos/M4, whose  
742 lowest seed is 0.98) and 0.020 (TimesFM at  $h=96$ ). So it is the stable end, not the drifting end,  
743 that would become threshold-sensitive if the cutoff were pushed past 0.975. We state the operating  
744 range explicitly rather than claiming threshold-independence: the conclusions hold for cutoffs in  
745  $[0.90, 0.975]$ .

746 Earlier versions of this table listed the Chronos/ETT cells at 0.886–0.949 and therefore reported them  
747 as *split* below 0.95. That range was misattributed—it belongs to the Moirai-Small sample-size sweep,  
748 not to Chronos—and the measured Chronos values are an order of magnitude lower, which removes  
749 the split.

750 **CKA (0.95).** Observed CKA clusters away from the cutoff at the drifting end: drifting cells sit at  
751 0.087–0.179, 0.19–0.64, 0.25–0.33, and 0.456; stable cells at 0.98–0.999. A random-initialization  
752 baseline gives CKA  $0.000 \pm 0.000$  (5 seeds), fixing the floor of the scale, so the Chronos/ETT cells at  
753 0.087–0.179 are much nearer that floor than the threshold. This is also why the Chronos/ETTh1-vs-  
754 ETTh2 comparison cannot be resolved by CKA at any cutoff: both cells are unambiguously drifted  
755 on every reading, yet the less-drifted of the two is the one that loses more to freezing.